

# Mathematical Analysis 2

## Seminar 8

Special functions

$$\boxed{\Gamma 1} \quad \Gamma(a) = \int_0^{\infty} e^{-x} x^{a-1} dx \quad (a > 0) \quad [\text{Def. 1}]$$

$$\boxed{\Gamma 2} \quad \Gamma(a+1) = a \cdot \Gamma(a) \quad [\diamond 6]$$

$$\boxed{\Gamma 3} \quad \Gamma(n+1) = n! \quad (n \in \mathbb{N}) \quad \Gamma(n) = (n-1)! \quad [\diamond 7]$$

$$\boxed{\Gamma 4} \quad \Gamma(a) \cdot \Gamma(1-a) = \frac{\pi}{\sin \pi a} \quad (a \in (0, 1)) \quad [\diamond 11]$$

$$\boxed{\Gamma 5} \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \quad [\diamond 12]$$

---


$$\boxed{\text{Ex. 1}} \quad \Gamma\left(\frac{3}{2}\right) \quad \Gamma\left(\underbrace{\frac{1}{2}+1}_a\right) \stackrel{(\Gamma 2)}{=} \frac{1}{2} \Gamma\left(\frac{1}{2}\right) \stackrel{(\Gamma 5)}{=} \frac{\sqrt{\pi}}{2}$$

$$\boxed{\text{Ex. 2}} \quad \Gamma(a+n) = a^{\bar{n}} \cdot \Gamma(a) \quad (n \in \mathbb{N})$$

$$\bar{a}^n = a(a+1) \cdots (a+n-1)$$

$$a^{\bar{0}} = 1$$

$$\begin{aligned} \Gamma(a+n) &\stackrel{(\Gamma 2)}{=} (a+n-1) \Gamma(a+n-1) \stackrel{(\Gamma 2)}{=} (a+n-1) \cdot (a+n-2) \Gamma(a+n-2) \\ &= \dots = \underbrace{(a+n-1)(a+n-2) \cdots a}_{a^{\bar{n}}} \Gamma(a) \end{aligned}$$

$$\textcircled{!} \quad \Gamma(\underbrace{n+1}) = 1^{\bar{n}} \underbrace{\Gamma(1)}_1 \text{ (from Def. 1)} \quad ; \quad 1^{\bar{n}} = 1 \cdot 2 \cdot 3 \cdots n = n!$$

$$\boxed{\text{Ex. 3}} \quad \Gamma\left(n + \frac{1}{2}\right) \stackrel{\text{Ex 2}}{=} \left(\frac{1}{2}\right)^{\bar{n}} \Gamma\left(\frac{1}{2}\right) \stackrel{(\Gamma 5)}{=} \frac{(2n-1)!!}{2^n} \sqrt{\pi}.$$

$$\left(\frac{1}{2}\right)^{\bar{n}} = \frac{1}{2} \cdot \left(\frac{1}{2} + 1\right) \cdots \left(\frac{1}{2} + n - 1\right) = \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{5}{2} \cdots \frac{2n-1}{2} = \frac{(2n-1)!!}{2^n} \quad \left((-1)!! \stackrel{\text{def}}{=} 1\right)$$

$$B1 \quad B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx \quad (a, b > 0) \quad [\text{Def. 2}]$$

$$B2 \quad B(a, b) = \int_0^\infty \frac{t^{a-1}}{(1+t)^{a+b}} dt \quad \leftarrow \begin{array}{l} x = \frac{t}{1+t} \\ x = 1-t \end{array}$$

$$B3 \quad B(a, b) = B(b, a)$$

$$B4 \quad B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad \leftarrow \begin{array}{l} \text{For } \Gamma(a+n), \text{ use Ex 2} \\ \text{Use also } (\Gamma^3) \end{array}$$

$$B5 \quad B(a, 1-a) = \frac{\pi}{\sin \pi a} \quad (a \in (0, 1))$$

$$B6 \quad B\left(\frac{1}{2}, \frac{1}{2}\right) = \pi \quad \leftarrow a = \frac{1}{2} \text{ or from (B1)}$$

$$B7 \quad B(a, n) = \frac{(n-1)!}{a(a+1)\cdots(a+n-1)} = \frac{(n-1)!}{a^n} \quad (n \in \mathbb{N}^*) \quad \leftarrow$$

[♦2]

[♦1]

[♦10]

[♦11]

[♦5]

[♦14]

**Ex. 4**

$$B(a+n, b+m) = \frac{a^{\bar{n}} \cdot b^{\bar{m}}}{(a+b)^{\overline{m+n}}} \cdot B(a, b) \quad (m, n \in \mathbb{N})$$

$$\underline{B(a+n, b+m)} \stackrel{(B4)}{=} \frac{\Gamma(a+n) \Gamma(b+m)}{\Gamma(a+b+n+m)} \stackrel{(Ex2)}{=} \frac{a^{\bar{n}} \Gamma(a) \cdot b^{\bar{m}} \Gamma(b)}{(a+b)^{\overline{m+n}} \Gamma(a+b)} = \frac{a^{\bar{n}} \cdot b^{\bar{m}}}{(a+b)^{\overline{m+n}}} \cdot \underbrace{\frac{\Gamma(a) \Gamma(b)}{\Gamma(a+b)}}_{B(a, b)}$$

$$\text{Ex. 5} \quad B(a+n, (1-a)+m) = \frac{a^{\bar{n}} \cdot (1-a)^{\bar{m}}}{(m+n)!} \cdot \frac{\pi}{\sin \pi a} \quad (m, n \in \mathbb{N}, a \in (0, 1))$$

$$\stackrel{\text{Ex 4}}{=} \frac{a^{\bar{n}} \cdot (1-a)^{\bar{m}}}{1^{\overline{m+n}}} B(a, 1-a) \stackrel{(B5)}{=}$$

**Some partial conclusion:**  $\Gamma$  and  $B$  have explicit values for:

$$\triangleright \Gamma(n), \quad n \in \mathbb{N}^*$$

$$\triangleright \Gamma\left(n + \frac{1}{2}\right), \quad n \in \mathbb{N}$$

$$\triangleright B(a, b), \quad \text{with } a \in \mathbb{N}^*, \text{ or } b \in \mathbb{N}^*, \text{ or } a + b \in \mathbb{N}^*$$

**Ex. 6**  $\int_0^1 (x-x^2)^n dx \quad (n \in \mathbb{N})$

$$= \int_0^1 x^n (1-x)^n dx = B(n+1, n+1) = \frac{\Gamma(n+1) \Gamma(n+1)}{\Gamma(2n+2)} = \frac{(n!)^2}{(2n+1)!}$$

**Ex. 7**  $\int_a^b (x-a)^{\frac{5}{2}} (b-x)^{\frac{9}{2}} dx \quad (a < b)$

If  $a=0, b=1 \Rightarrow I = \int_0^1 x^{5/2} (1-x)^{9/2} dx = B\left(\frac{7}{2}, \frac{11}{2}\right) = \frac{\Gamma(\frac{7}{2}) \Gamma(\frac{11}{2})}{\Gamma(9)} = \frac{\cancel{3} \cdot \cancel{5} \cdot \cancel{3} \cdot \cancel{5} \cdot \cancel{7} \cdot 9}{\cancel{2}^8 \cdot \cancel{2} \cdot \cancel{3} \cdot \cancel{4} \cdot \cancel{5} \cdot \cancel{6} \cdot \cancel{7} \cdot \cancel{8}} \cdot \pi$

$\Gamma(\frac{7}{2}) = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma(\frac{1}{2}) = \frac{15}{8} \sqrt{\pi} = \frac{3 \cdot 5}{2^3} \sqrt{\pi}$

$\Gamma(9) = 8!$

$\Gamma(\frac{11}{2}) = \frac{9}{2} \cdot \frac{7}{2} \cdot \Gamma(\frac{7}{2}) = \frac{7 \cdot 9 \cdot 15}{2^5} \sqrt{\pi} = \frac{3 \cdot 5 \cdot 7 \cdot 9}{2^5} \sqrt{\pi}$

$= \frac{45}{2^{15}} \pi$

In general:  $[a, b] \rightarrow [0, 1]$

change of variable:

$x = pt + q \quad \begin{cases} x=a \Leftrightarrow t=0 \\ x=b \Leftrightarrow t=1 \end{cases} \Leftrightarrow \begin{cases} q=a \\ p+q=b \end{cases} \Leftrightarrow \begin{cases} p=b-a \\ q=a \end{cases}$

$x = (b-a)t + a$

$dx = (b-a)dt; \quad x-a = (b-a)t; \quad b-x = (b-a)(1-t)$

$I = \int_0^1 (b-a)^{5/2} t^{5/2} (b-a)^{9/2} (1-t)^{9/2} (b-a) dt = (b-a)^8 \int_0^1 t^{5/2} (1-t)^{9/2} dt$

$(b-a)^8 \cdot \frac{45}{2^{15}} \pi$

**Ex. 8**  $\int_0^{\frac{\pi}{2}} \sin^p x \cdot \cos^q x \, dx \quad (p, q > -1)$

$$B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt ; \underline{t = \sin^2 x} ; 1-t = \cos^2 x ; dt = \underline{2 \sin x \cdot \cos x \, dx}$$

$$B(a, b) = 2 \int_0^{\pi/2} \sin^{2a-1} x \cdot \cos^{2b-1} x \, dx \quad \begin{cases} 2a-1 = p \\ 2b-1 = q \end{cases} \Rightarrow \begin{cases} a = \frac{p+1}{2} \\ b = \frac{q+1}{2} \end{cases}$$

$$I = \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right).$$

**Ex. 9**  $\int_0^{\frac{\pi}{2}} \tan^p x \, dx \quad (p \in (-1, 1)) = \int_0^{\pi/2} \sin^p x \cdot \cos^{-p} x \, dx \stackrel{\text{Ex. 8}}{=}$

$$= \frac{1}{2} B\left(\underbrace{\frac{p+1}{2}}_a, \underbrace{\frac{-p+1}{2}}_b\right) = \frac{1}{2} B(a, 1-a) \stackrel{\uparrow}{=} \frac{\pi}{2 \cdot \sin \pi a} = \frac{\pi}{2 \cdot \sin \pi \frac{p+1}{2}} =$$

$$(a+b=1) \quad = \frac{\pi}{2 \cdot \cos \frac{\pi p}{2}}$$



**Ex. 10**

$$\int_0^{\infty} \frac{x^p}{(1+x^q)^r} dx \quad (p > -1, \quad q > 0, \quad \underline{rq - p > 1}) \quad \parallel \quad B(a, b) = \int_0^{\infty} \frac{t^{a-1}}{(1+t)^{a+b}} dt$$

$$t = x^q; \quad x = t^{1/q}; \quad dx = \frac{1}{q} t^{\frac{1}{q}-1} dt \Rightarrow I = \frac{1}{q} \int_0^{\infty} \frac{t^{p/q + 1/q - 1}}{(1+t)^r} dt$$

$$\begin{cases} a-1 = \frac{p+1}{q} - 1 \Rightarrow a = \frac{p+1}{q} \\ a+b = r \Rightarrow b = r-a \Rightarrow b = r - \frac{p+1}{q} \end{cases}$$

$$I = \frac{1}{q} B\left(\frac{p+1}{q}, r - \frac{p+1}{q}\right)$$

When is  $I$  an explicit value?

$$\frac{p+1}{q} \in \mathbb{N}^*, \text{ or } r - \frac{p+1}{q} \in \mathbb{N}^*, \text{ or } r \in \mathbb{N}^*$$

**Ex. 11**

$$\int_0^{\infty} \frac{x^p}{x^2+1} dx \quad (p \in (-1, 1))$$

**Ex. 12**

$$\int_0^{\infty} \frac{x^3}{1+x^4 \cdot \sqrt[4]{x}} dx$$